

Phase 2:Requirement Analysis

SmartLenderAI-Powered Credit Card Approval System

Role	Name
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Functional Requirements

ID	Requirement
FR-1	Accept applicant details through a web form (income, employment, credit history, etc.)
FR-2	Preprocess and encode inputs the same way as the training pipeline
FR-3	Return an Approve / Reject prediction with a confidence score
FR-4	Allow retraining the model on updated historical data
FR-5	Support cloud deployment through IBM Watson Machine Learning

Non-Functional Requirements

ID	Requirement
NFR-1	Prediction response time under 2 seconds
NFR-2	Web form usable without technical training
NFR-3	Model retraining reproducible from raw data to deployment
NFR-4	System available for real-time, concurrent predictions

Data Requirements

Historical applicant records with demographic and financial fields (income, employment length, existing credit, education, family status) are used together with a payment history log that is used to derive the approval label.

Hardware & Software Requirements

- Python 3.10+, pip
- 8 GB RAM minimum for local training
- Web browser for the Flask front end
- IBM Cloud account (only required for Watson ML deployment)